

AATC Facility Requirements Document

This document provides details about minimum facility requirements for training centers broken out by Apple courses.

- The Training Center is responsible for maintaining the Facility Requirements and is responsible for providing all hardware and software described in this document; for each classroom, network, and projection system setup; and for installing the software prior to the first day of class.
- The Certified Trainer must have access to the classroom at least one business day prior to the first day of class. It is recommended that throughout the Apple Course, the Certified Trainer have easy access to the classroom, including after-hours and weekends.

Minimum Classroom Recommendations for any Apple course taught –

- Minimum recommended classroom size is 40-feet by 40-feet.
- Student work surfaces should be at least 36 inches wide by 28 inches deep, to accommodate two student systems.
- The instructor's table should be at least 48 inches wide by 28 inches deep, and located at the front of the classroom.
- Armchairs are recommended for all students and the instructor. Padded seats are preferred.
- Power outlets need to be provided for all computers, monitors, and peripheral devices used in the class, and should provide approximately five (5) amps of power for each device.
- The projection screen at the front of the classroom should be at least seven (7) feet away from the nearest edge of the student tables.
- Lighting controls should be accessible by the instructor. Dimming capability is preferred.
- Temperature controls should be accessible by the instructor.
- A wall-mounted or freestanding white board (with dry-erase markers and erasers) is suggested and should be located at the front of the classroom.
- A Mac OS X-compatible USB raster printer (with appropriate interface cable) with 10 foot cord is needed and should be accessible to the instructor.
- The classroom should be free of noisy machinery so the instructor can converse with students comfortably.
- Phones and restrooms should be easily accessible.
- One 100BASE-T Ethernet switch with $n+5$ or more ports (where n =number of students)
- Projection system (as defined below)

Minimum Projection System Requirements

- The projection equipment needs to be on a sturdy table or stand (approximately 29 inches high), or ceiling mounted.
- Minimum requirements for the projection system are as follows:
 - Resolution of 1024 by 768 pixels @ 75Hz
 - Projection rating of at least 2,000 lumens
 - 6 x 6 foot or larger projection screen
 - Spare projector bulbs
 - VGA switchbox

- 3-port Switchbox capable of allowing projection from either Instructor System
- Appropriate cables to connect the switchbox to the projection system
- Cabling to simultaneously connect two monitors to the switchbox

Minimum Network Requirements

- Each student station should be isolated from any other customer or classroom network, on a separate 100-Mbps Ethernet switch with $n+5$ or more ports (where n =number of students).
- Network must be isolated from any other customer network.
- All machines should be networked and have their connectivity tested prior to the first day of class
- Internet access or local subnet setup for all systems
- **Optional** - Airport

Minimum Student System Requirements

- Each student station should consist of one computer system, a client system, along with a 4-port 100-Mbit hub.
- One dedicated student station must be provided for each student in the class. Because systems will be re-imaged before the start of class, provide only systems whose hard drives can be erased.
- Macintosh computer with PowerPC G3, G4, or G5 processor with built-in USB
- Display with 1024 by 768 pixels, 16-bit resolution
- 128 MB of physical RAM (256MB RAM preferred)
- 2GB available hard disk space (4GB hard drive preferred)
- DVD-ROM
- 100BASE-T Ethernet
- **Optional** – Airport card

Minimum Instructor System Requirements

- Each instructor station should consist of two (2) computer systems; a client/presentation system and a server system, along with a 4-port 100-Mbit hub.
- One dedicated instructor station must be provided for each classroom. Because systems will be re-imaged before the start of class, provide only systems whose hard drives can be erased.
- Macintosh computer with PowerPC G3, G4, or G5 processor with built in USB
- Display with 1024 by 768 pixels, 16-bit resolution
- 128 MB of physical RAM (256MB RAM preferred)
- 2GB available hard disk space (4GB hard drive preferred)
- DVD-ROM
- 100BASE-T Ethernet
- **Optional** – Airport card

Minimum Software Requirements for both Students and Instructor

- One copy of Mac OS X v10.3 installed (use clean default easy install), including NetBoot, is required for each student and Instructor station for all Mac OS X courses.
- It is the responsibility of the Training Center to ensure valid software licenses from Apple or any 3rd party vendor for all computer systems used by Training Center for students and Instructors participating in the specific Apple Course, including updates that may be required.

Mac OS X Fundamentals and Mac OS X Daily Productivity

Same as above requirements

Creating Presentations with Keynote

Minimum Software Requirements for both Students and Instructor- All minimum requirements listed above, plus:

- One copy of Keynote 1.1 installed is required for each student and Instructor station for all Mac OS X courses.

Minimum Instructor System Requirements - All minimum requirements listed above, plus:

- Video output for connection to projector

Mac OS X Help Desk Essentials

Minimum Classroom Recommendations for any Apple course taught – All minimum requirements listed above, plus:

- Eight FireWire cables (6-pin to 6-pin, male to male, 3 feet long).
- **Optional** – Six USB Bluetooth transceivers

Minimum Student System Requirements- All minimum requirements listed above, plus:

- Built-in FireWire port

Minimum Instructor Systems Requirements- All minimum requirements listed above, plus:

The instructor systems include one machine from which to display the slides and demonstrate Mac OS X and one machine, which acts as a classroom server to demonstrate Mac OS X Server.

Instructor System 1

- Macintosh v10.3-compliant G4 or G5 computer
- Video output for connection to projector
- 256MB RAM
- 4GB hard drive
- AirPort Card installed
- Network access to both classroom network and Internet

Instructor System 2

- Macintosh v10.3-compliant G4 or G5 computer
- Video output for connection to projector
- 256MB RAM
- 4GB hard drive

Mac OS X Server Essentials

Minimum Classroom Recommendations for any Apple course taught - All minimum requirements listed above, plus:

- **Optional** - One Xserve

Minimum Student Systems Requirements- All minimum requirements listed above, plus:
The student workspace consists of two computer systems and a 100-MBPS hub.

IMPORTANT Apple Training supplies Mac OS X Server v10.3 installations CDs for use in this class. However, the AATC must supply the installation discs that originally shipped with the machines.

Student System 1

- Xserve or Power Mac G4 or G5 computer
- 256MB RAM
- 8GB hard drive
- 100-Mbps Ethernet

Student System 2

- FireWire port
- 100-Mbps Ethernet

Minimum Instructor Systems Requirements- All minimum requirements listed above, plus:
The instructor systems include one machine from which to display the slides and demonstrate Mac OS X and one machine, which acts as a classroom server to demonstrate Mac OS X Server.

Instructor System 1

- Xserve or Power Mac G4 or G5 computer
- Video output for connection to projector
- 256MB RAM
- 6GB hard drive
- 100-Mbps Ethernet

Instructor System 2

- 256MB RAM
- 4GB hard drive
- Video output for connection to projector
- 100-Mbps Ethernet
- Keynote

System Administration of Mac OS X Clients

Minimum Classroom Recommendations for any Apple course taught - All minimum requirements listed above, plus:

- Sixteen (16) FireWire cables (6-pin to 6-pin, male to male, 3 feet long).

Minimum Network Requirements- All minimum requirements listed above, plus:

- One 100-Mbps Ethernet switch environment to connect all machines together is required. This network should be isolated from any other network and should have enough connections for three instructor machines and all student machines.

Minimum Student System Requirements- All minimum requirements listed above, plus:

Student System 1

- Xserve or Power Mac G4 or G5 computer
- 256MB RAM

- 8GB hard drive
- 100-Mbps Ethernet
- Mac OS X Server v10.3 (standard installation; updated to version 10.3.3)

Student System 2

- FireWire port
- 100-Mbps Ethernet
- Mac OS X Server v10.3 (standard installation; updated to version 10.3.3)

Minimum Instructor Systems Requirements- All minimum requirements listed above, plus:
The instructor systems include one machine from which to display the slides and demonstrate Mac OS X, one machine, which acts as a classroom server to demonstrate Mac OS X Server and one machine, which acts as a PC server.

Instructor System 1

- Power Mac G4 or G5 computer
- Video output for connection to projector
- 256MB RAM
- 4GB hard drive
- 100-Mbps Ethernet
- Mac OS X Server v10.3 (default installation)

Instructor System 2

- Power Mac G4 or G5 computer
- Video output for connection to projector
- 256MB RAM
- 4GB hard drive
- 100-Mbps Ethernet

Instructor System 3

- Certified system for Windows 2000 Server
- 256MB RAM
- 4GB hard drive
- VGA output
- CD-ROM capable drive
- 100-Mbps Ethernet
- Windows 2000 Server with Service Pack 3
- Display supporting 1024 by 768 pixels, 16-bit color

System Administration Using Mac OS X Server

Minimum Classroom Recommendations for any Apple course taught - All minimum requirements listed above, plus:

- Sixteen (16) FireWire cables (6-pin to 6-pin, male to male, 3 feet long).
- One display switchbox must be available, allowing the Windows, Power Mac, and presentation system to share the projection system.

Minimum Network Requirements- All minimum requirements listed above, plus:

- One 100-Mbps Ethernet switch environment to connect all machines together is required. This network should be isolated from any other network and should have enough connections for three instructor machines and all student machines.

Minimum Student System Requirements- All minimum requirements listed above, plus:

- Built-in FireWire port

Minimum Instructor Systems Requirements- All minimum requirements listed above, plus:

The instructor systems include one machine from which to display the slides and demonstrate Mac OS X and one machine, which acts as a classroom server to demonstrate Mac OS X Server.

Instructor System 1

- Power Mac G4 or G5 computer
- Video output for connection to projector
- 256MB RAM
- 4GB hard drive
- 100-Mbps Ethernet
- Mac OS X Server v10.3 (unlimited license; updated to version 10.3.3)

Instructor System 2

- Power Mac G4 or G5 computer
- Video output for connection to projector
- 256MB RAM
- 4GB hard drive
- 100-Mbps Ethernet
- Mac OS X v10.3 (erase install – no configuration; updated to version 10.3.3)

Directory Services

Minimum Classroom Recommendations for any Apple course taught - All minimum requirements listed above, plus:

- Sixteen (16) FireWire cables (6-pin to 6-pin, male to male, 3 feet long).
- One display switchbox must be available, allowing the Windows, Power Mac, and presentation system to share the projection system.

Minimum Network Requirements- All minimum requirements listed above, plus:

- One 100-Mbps Ethernet switch environment to connect all machines together is required. This network should be isolated from any other network and should have enough connections for three instructor machines and all student machines.

Minimum Student System Requirements- All minimum requirements listed above, plus:

Student System 1

- Xserve or Power Mac G4 or G5 computer
- 256MB RAM
- 8GB hard drive
- 100-Mbps Ethernet

- Mac OS X Server v10.3 (standard installation; updated to version 10.3.3)

Student System 2

- FireWire port
- 100-Mbps Ethernet
- Mac OS X Server v10.3 (standard installation; updated to version 10.3.3)

Minimum Instructor Systems Requirements- All minimum requirements listed above, plus:
The instructor systems include one machine from which to display the slides and demonstrate Mac OS X, one machine, which acts as a classroom server to demonstrate Mac OS X Server and one machine, which acts as a PC server.

Instructor System 1

- Power Mac G4 or G5 computer
- Video output for connection to projector
- 256MB RAM
- 4GB hard drive
- 100-Mbps Ethernet
- Mac OS X Server v10.3 (unlimited license; updated to version 10.3.3)

Instructor System 2

- Power Mac G4 or G5 computer
- Video output for connection to projector
- 256MB RAM
- 4GB hard drive
- 100-Mbps Ethernet
- Mac OS X v10.3 (erase install – no configuration; updated to version 10.3.3)

Instructor System 3

- Certified system for Windows 2000 Server
- 256MB RAM
- 4GB hard drive
- Video output for connection to projector
- CD-ROM capable drive
- 100-Mbps Ethernet
- Windows 2000 Server with Service Pack 3 (with enough client licenses to support all students)
- Display supporting 1024 by 768 pixels, 16-bit color

Internet Application Development Using WebObjects

Minimum Student System Requirements- All minimum requirements listed above, plus:

- Power Mac G4 or G5
- 256MB RAM
- 10GB hard drive
- 100-Mbps Ethernet
- Mac OS X v10.3 (full installation, updated to version 10.3.3)
- Mac OS X Developer Tools 1.1 or later (installed)
- WebObjects 5.2.3 (full installation)

Minimum Instructor Systems Requirements- All minimum requirements listed above, plus:
Refer to <http://docs.info.apple.com/article.html?artnum=107649> for installation instructions, if you are creating a new Mac OS X WebObjects 5.2 system.

Instructor System

- Power Mac G4 or G5 computer
- Video output for connection to projector
- 256MB RAM
- 10GB hard drive
- 100-Mbps Ethernet
- Mac OS X v10.3 (full installation, updated to version 10.3.3)
- Mac OS X Developer Tools 1.1 or later (installed)
- WebObjects 5.2.3 (full installation)
- Keynote (provided by instructor)

Advanced WebObjects Development

Minimum Student System Requirements- All minimum requirements listed above, plus:

- Power Mac G4 or G5
- 256MB RAM
- 10GB hard drive
- 100-Mbps Ethernet
- Mac OS X v10.3 (full installation, updated to version 10.3.3)
- Mac OS X Developer Tools 1.1 or later (installed)
- WebObjects 5.2.3 (full installation)

Minimum Instructor Systems Requirements- All minimum requirements listed above, plus:
Refer to <http://docs.info.apple.com/article.html?artnum=107649> for installation instructions, if you are creating a new Mac OS X WebObjects 5.2 system.

Instructor System

- Power Mac G4 or G5 computer
- Video output for connection to projector
- 256MB RAM
- 10GB hard drive
- 100-Mbps Ethernet
- Mac OS X v10.3 (full installation, updated to version 10.3.3)
- Mac OS X Developer Tools 1.1 or later (installed)
- WebObjects 5.2.3 (full installation)
- Keynote (provided by instructor)

Developing Desktop Applications With Cocoa

Minimum Student System Requirements- All minimum requirements listed above, plus:

- 256MB RAM
- 10GB hard drive
- 100-Mbps Ethernet
- Mac OS X v10.3 (full installation, updated to version 10.3.3)
- Mac OS X Developer Tools 1.1 or later (installed)

**Minimum Instructor Systems Requirements- All minimum requirements listed above, plus:
Instructor System**

- Power Mac G4 or G5 computer
- Video output for connection to projector
- 256MB RAM
- 40GB hard drive
- 100-Mbps Ethernet
- Mac OS X v10.3 (full installation, updated to version 10.3.3)
- Mac OS X Developer Tools (December 2002 or later)